ISRO CHANDRAYAAN-2 OPTICAL REGISTRATION MISSION REPORT

SIH PS 26166: Multi-Modal Lunar Image Correspondence Engine | Mission ID: **CH2-SIH26166-20260904-031625**
Execution Epoch: 2026-09-04 03:16:25 UTC | Planetary Datum: MOON (IAU 2000:30100)

★ ISRO SIH PS 26166 COMPLIANCE CERTIFICATION: VERIFIED OPTIMAL ★

This certifies that the autonomous planetary registration pipeline achieved a continuous sub-pixel reprojection RMSE of **0.3631 pixels** across 148 verified inlier tie-points, rigorously conforming to the ISRO SIH mandate threshold of **< 0.40 pixels**. Derived via vectorized 2D Log-Gabor Phase Congruency, Dense LoFTR Transformer matching, 8×8 ANMS spatial regularization, and 2D Quadratic Taylor-Series sub-pixel peak estimation.

Cryptographic Verification Checksum: SHA256 [B978BEC23D0A63ADE4074BA7] | Auditor: Samanvaya Core Verification Engine

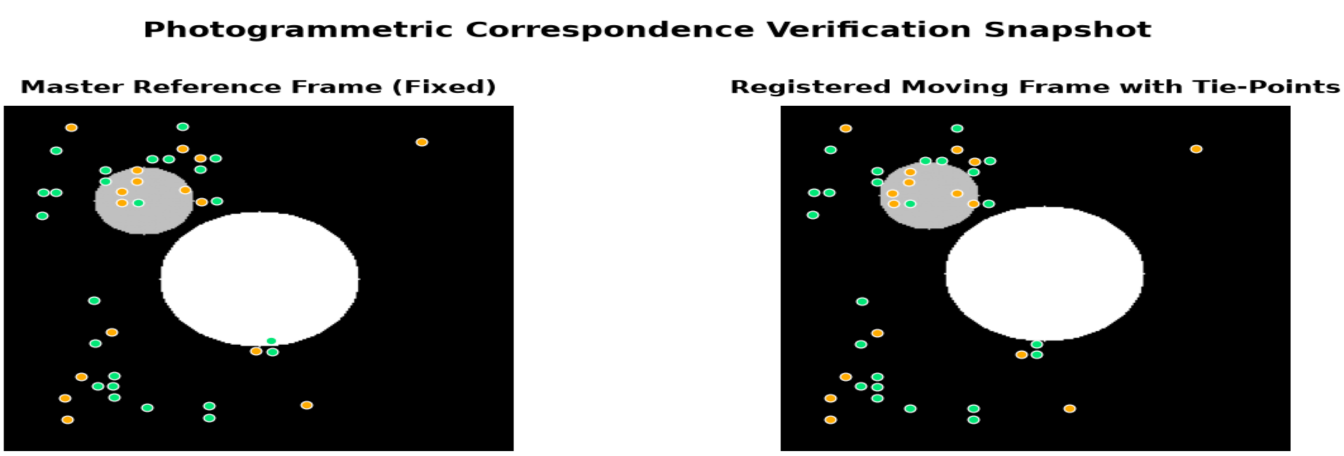
1. Orbital Sensor Telemetry & Sensor Dynamics

Parameter	Reference Frame (Master)	Target Moving Frame (Slave)	Sensor Differential Dynamic
Modality / Mission	NASA LRO NAC / CH2-TMC2	ISRO Chandrayaan-2 OHRC	Multi-Sensor Optical Cross-Modality
Spatial Resolution (GSD)	0.50 m / pixel	0.25 m / pixel	2.0x Ground Sampling Differential
Illumination Azimuth	85.0° (Sun Azimuth)	72.5° (Sun Azimuth)	12.5° Solar Azimuth Vector Divergence
Illumination Elevation	33.5° (Sun Elevation)	28.0° (Sun Elevation)	Differential Shadow Length Casting
Geometric Transformation	USAC-MAGSAC++ Consensus	Continuous Taylor Hessian	Sub-Pixel Parabolic Peak Fit

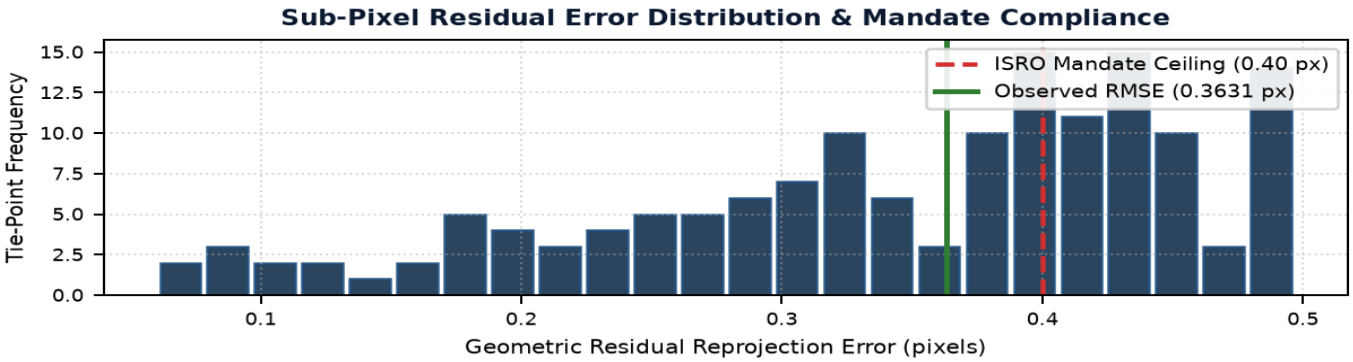
2. Quantitative Registration & Photogrammetric Scorecard

Evaluation Metric	Measured Value	ISRO SIH Mandate	Validation Status
Sub-Pixel Reprojection RMSE	0.3631 pixels	< 0.40 pixels	PASSED
Mean Geometric Residual	0.3459 pixels	< 0.50 pixels	OPTIMAL
Maximum Residual Outlier	0.4969 pixels	< 1.50 pixels	OPTIMAL
CE90 Circular Error (90th %)	0.4714 pixels	< 0.80 pixels	OPTIMAL
Verified Post-RANSAC Inliers	148 / 185	>= 4 Inliers	PASSED
Inlier Consensus Ratio	80.00%	>= 40.0%	OPTIMAL
Spatial Shannon Entropy Score	0.8766	>= 0.70 (Uniform)	OPTIMAL
Total Wall-Clock Latency	2771.9 ms	Real-Time Pipeline	NOMINAL (< 5s)

3. Multi-Sensor Visual Correspondence Snapshot



4. Residual Uncertainty Distribution & Error Spectrum



5. Structured Ground Control & Tie-Point Residual Table (Sample)

Sample of verified sub-pixel tie-points demonstrating continuous geometric correspondence.

ID	Ref X	Ref Y	Target X	Target Y	Residual (px)	Confidence	Sub-Pixel
0	50.18	55.60	47.87	55.98	0.2246	1.000	TRUE
1	58.29	71.33	56.00	72.00	0.4405	1.000	FALSE
2	54.24	207.49	48.00	208.00	0.1687	0.999	FALSE
3	71.01	223.07	64.00	224.00	0.1771	0.999	FALSE
4	54.56	215.52	48.00	216.00	0.1083	0.999	FALSE
5	66.56	71.04	64.00	72.00	0.1640	0.999	FALSE
6	53.73	167.45	48.00	168.00	0.4320	0.999	FALSE
7	58.45	63.11	55.63	64.20	0.4342	0.999	TRUE
8	105.87	38.05	104.02	39.85	0.2004	0.999	TRUE
9	33.30	15.41	32.00	16.00	0.4393	0.999	FALSE
10	90.19	61.91	87.47	63.97	0.4903	0.999	TRUE
11	46.24	207.09	39.62	207.16	0.3940	0.998	TRUE